

ENGINEERING CALCULATION REPORT · AE-AL-2026-001 · REV A · PRELIMINARY

Rod-pump surface-card analysis

Why a 4,200 ft rod string produces load undulations on the surface card, what the card confirms about the well, and what is required to progress to a downhole pump diagnosis. **Preliminary results** — kinematics and card integrity are complete; pump diagnosis is pending data.

4.56_{predicted} Undulations per half stroke — 4 observed	1.03_s Load-peak interval, 60/N_o'
38.8_{bfpd} Displacement on plunger stroke	2,412_{lb} Unexplained load datum excess

1 Objective

Purpose and scope of this calculation.

OBJECTIVE

Quantify the rod-string free-vibration response visible on the surface dynamometer card, verify the card against first principles, and identify the data required to progress to a downhole pump diagnosis.

Scope: surface-card kinematics and card integrity

Method: API RP 11L; SPE 18189

Status: preliminary

These are preliminary results. The kinematic and card-metric findings are complete and stand on their own. The pump diagnosis does not: it requires inputs not yet available, and the diagnostic classifier behind it has not been calibrated against field cards with known outcomes. Sections are colour-coded by provenance throughout.

2 Design data

Inputs as supplied, and assumptions where a value was not.

PUMPING UNIT	
Surface stroke <i>S</i>	41 in
Pumping speed <i>N</i>	6.4 SPM
Prime mover	Arrow C-66 (13 hp)
Gearbox rating	not supplied

DOWNHOLE	
Pump setting depth D	4,300 ft
Plunger diameter	1.25 in
Pump type	top-hold-down insert
Rod string L	4,200 ft of 3/4 in steel
Taper	single, $F_c = 1.000$

OPERATING	
Tubing pressure	150 psi
Casing pressure	25 psi
Production, previous day	23 oil / 0 water bbl
Runtime	not supplied h/d

2.1 Assumptions carried

Each of the following was assumed, not measured. Results depending on them are marked accordingly.

ASSUMED VALUES	
Fluid specific gravity	0.85 -
Fluid level above pump	4,300 (submerged) ft
Plunger overtravel	0 (conservative) in
Formation volume factor B_o	unknown -

CARD DIGITISATION

Source	screenshot from thread
Load resolution	± 40 lb
Position resolution	± 0.3 in
Samples	80 per branch

3 Analysis methodology

Governing relations, shown explicitly so a reviewer can check the arithmetic by hand.

3.1 Rod-string free vibration

The rod string is an elastic bar fixed at the pump and free at the polished rod. It rings at its own natural frequency, independently of pumping speed.

$$N_o = \frac{245,000}{L} \quad (1)$$

N_o Undamped natural frequency SPM L Rod string length ft

The constant 245,000 is $15c$ for steel. Computing from wave speed directly gives 58.21 SPM against 58.33 SPM from the API constant — 0.20% agreement, which the implementation asserts rather than assumes. A non-steel rod trips that check.

$$\Delta t = \frac{60}{N_o'} \quad (2)$$

Δt	Interval between load peaks	s	N_o'	Taper-adjusted natural frequency	SPM
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$$n = \frac{0.5}{N / N_o'} \quad (3)$$

n	Undulations per half stroke	-	N	Pumping speed	SPM
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3.2 Loads and plunger travel

$$F_o = 0.433 SG D A_p + (P_t - P_c) A_p \quad (4)$$

F_o	Fluid load on plunger	lb	A_p	Plunger area	in ²
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$$K_r = \frac{1}{E_r L}, E_r = \frac{12}{A_r E} \quad (5)$$

K_r	Rod string spring rate	lb/in	E_r	Rod elastic constant	in/lb/ft
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$$S_p = S - \frac{F_o}{K_r} + S_{ot} \quad (6)$$

S_p	Plunger stroke	in	S_{ot}	Overtravel	in
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Rod stretch under fluid load shortens plunger travel. This is the step most often skipped, and skipping it biases displacement high.

$$PD = 0.1484 A_p S_p N \quad (7)$$

PD	Theoretical pump displacement	bfpd
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3.3 Surface-to-downhole conversion

The downhole card is obtained by marching the damped wave equation down the rod string and rebuilding load from the strain field. Load is never inherited from the surface: an affine rescale would preserve every vibration harmonic and could not diagnose anything.

$$F = EA \frac{\partial u}{\partial x}$$

(8)

F

Load at depth

lb

u

Axial displacement field

in

EA

Local axial stiffness

lb

4 Results — preliminary

Each result carries the confidence it can support.

4.1 Rod-string response — the undulation question

COMPUTED	
Natural frequency N_o	58.33 SPM
Peak interval Δt	1.029 s
Speed ratio N/N_o'	0.110 -
Predicted undulations n	4.56 per half stroke

OBSERVED ON CARD	
Clear load humps, upstroke	4 + 1 partial
Agreement	within digitisation error
Regime	wave-dominated
Interpretation	expected behaviour

VALIDATED The undulations are the rod string ringing at its own frequency. They crowd toward the top of the card because polished-rod velocity approaches zero there, compressing a fixed time interval into a shorter distance — the card plots position, the ringing is periodic in time. At this speed their absence would be surprising.

4.2 Card metrics

LOADS	
PPRL	12,438 lb @ 7.5 in
MPRL	9,274 lb @ 35.5 in
Load range	3,164 lb
Rod weight in air W_r	6,863 lb

WORK AND POWER	
Card area	69,950 lb-in
Work per stroke	5,829 ft-lb
Polished-rod power	1.13 hp
Spring rate K_r	271.7 lb/in

VALIDATED Independent of every assumption in section 2.

4.3 Load datum anomaly — the card implies a heavier string

The minimum polished-rod load exceeds the weight of the entire rod string hanging in air by 2,412 lb. This has no mechanical explanation. On the downstroke friction acts upward, resisting the descending rods, which lowers polished-rod load — friction widens a card, it cannot lift one. Two explanations remain: a load cell reading high, or a string heavier than the 3/4 in reported. Testing each candidate rod size against both ends of the card separates them — a valid size must put the minimum load below the string's air weight, and the peak above static weight plus fluid load by a dynamic margin consistent with this well's speed.

MEASURED AGAINST CANDIDATE ROD SIZES	
3/4 in — MPRL vs air weight	+2,412 (impossible) 1b
3/4 in — PPRL vs static	+3,480 (implausible) 1b
7/8 in — MPRL vs air weight	-67 1b
7/8 in — PPRL vs static	+1,002 1b
1 in — PPRL vs static	-1,854 (impossible) 1b
CANDIDATE CAUSES – BOTH LIVE	
String heavier than reported	card is consistent with 7/8 in
Load-cell zero or scale offset	equally possible
Effect on load differences	none – offset cancels
Effect on absolute loads	not usable until resolved

ANALYTICAL 7/8 in is the only size where both ends of the card are physically consistent. At 6.4 SPM on a 4,200 ft string roughly 1,000 lb of dynamic load above static is expected, which 7/8 in gives; the 3,480 lb that 3/4 in would require is far beyond what this speed produces. The operator reported 3/4 in, so this is not asserted as fact — recommend checking the rod tally or last workover ticket alongside the load-cell calibration, since strings can gain heavier top joints without reaching the paperwork.

4.4 Ideal pump card — reference for the downhole comparison

The theoretical card a perfectly filled pump would produce at this setting. It is the reference the calculated downhole card will be measured against once Stage B data arrives.

IDEAL PUMP CARD	
Fluid load on plunger	1,942 lb
Peak / minimum load	1,942 / 0 lb
Card area	79,630 lb-in
Basis stroke	41 (surface) in

QUALIFICATIONS	
Hand check vs $0.433 \cdot SG \cdot D \cdot A_p$	agrees to 0.002 %
Domain	pump card, not surface
Built on surface stroke	overstates area ~22 %
Plunger stroke after stretch	~33.3 in

ANALYTICAL This is a **pump-domain** reference and must not be compared with the measured surface card area of 69,950 lb-in — they are different quantities, and their apparent closeness is coincidence. The ideal card is also built on the 41 in surface stroke rather than the ~33.3 in plunger stroke, overstating its area by roughly 22% on that basis alone. Shape-similarity and deviation metrics are omitted: they are computed against a non-monotonic reference and do not survive inspection.

4.5 Power consumption

MEASURED FROM THE CARD	
Work per stroke	5,829 ft-lb
Polished-rod power	1.131 hp
Prime-mover requirement	2.803 hp
Electrical equivalent	2.090 kW
ASSUMPTIONS CARRIED	
Cyclic load factor	1.897 (default)
Prime-mover efficiency	0.85 (default)
Pumping-unit efficiency	0.90 (default)
Daily energy at 8 / 12 / 16 h	16.7 / 25.1 / 33.4 kWh

ANALYTICAL Work per stroke and polished-rod power are read from the card and are invariant to a constant load offset, so they survive the datum question above. The prime-mover figure is weaker: the 1.897 cyclic load factor is drawn from a NEMA B *electric motor* table and applied to an Arrow C-66 *natural gas engine*, for which NEMA slip is undefined; selecting NEMA D instead gives 2.03 hp, a 27% swing from a default. Against the C-66's 13 hp rating this is a cycle-average load factor near 22%, which indicates **headroom to raise speed or stroke** if the reservoir supports it — not a sizing error, since peak crank torque is what sizes a gas engine and cannot be evaluated without the unit geometry. Daily energy is quoted as a rate because runtime is unknown.

4.6 Displacement and efficiency

CARRYING ASSUMED SG AND FLUID LEVEL	
Fluid load F_o	2,096 lb
Rod stretch	7.7 in
Plunger stroke S_p	33.3 in
Displacement PD	38.8 bfpd
SENSITIVITY	
SG 0.80–0.90, level 3,000–4,300 ft	38.3 – 41.7 bfpd
Spread	±4 %
Conclusion	assumptions not limiting
Volumetric efficiency	not determinable

ANALYTICAL Efficiency is not reported. A unit cycling on a pump-off controller produces reduced volume with a healthy pump, indistinguishable from low fillage on daily volume alone. Using the 41 in surface stroke instead of the 33.3 in plunger stroke would overstate displacement at 47.8 bfpd.

5 Validation status

Honest provenance for every headline claim.

Rod-string kinematics and undulation count

API RP 11L; two independent routes to N_o agree to 0.20%. Depends only on stroke, depth, speed and the card.

VALIDATED



Card metrics — PPRL, MPRL, area, polished-rod power

Read directly from the digitised card; independent of all assumptions.

VALIDATED



Surface-to-downhole solver

SPE 18189 implementation reproduces reference downhole cards at 0.9% median normalised RMSE, correlation 1.000, across five validation wells including deviated hole.

VALIDATED



Plunger stroke and theoretical displacement

Carries assumed fluid SG and level. Sensitivity sweep bounds the effect at $\pm 4\%$.

ANALYTICAL



Work per stroke and polished-rod power

Card area reproduced by two independent integration methods agreeing to 0.000%, and invariant to a constant load offset, so both survive the unresolved datum question.

VALIDATED



Ideal pump card — fluid load and card area

Fluid load matches the closed-form $0.433 \cdot SG \cdot D \cdot A_p$ to 0.002%. Carries the assumed fluid SG, and is built on surface rather than plunger stroke. Pump-domain only.

ANALYTICAL



Prime-mover power requirement

Cyclic load factor 1.897 comes from a NEMA B electric-motor table applied to a natural gas engine; efficiencies are library defaults. Selecting NEMA D instead moves the result 27%.

ANALYTICAL



Rod string size as run

Operator reported 4,200 ft of 3/4 in. The card is consistent with 7/8 in and inconsistent with 3/4 in at both ends. Requires the rod tally to resolve; not asserted either way.

PENDING



Absolute polished-rod loads

Suspended pending resolution of the 2,412 lb datum anomaly, which may be a load-cell offset or a heavier string than reported. Load differences remain usable.

PENDING



Card shape-similarity and deviation metrics

Omitted from this report. Computed against a non-monotonic reference and sign-clamped, so an inverted card scores identically to an unrelated one.

PENDING



Volumetric efficiency

Requires runtime hours per day and formation volume factor. Not assumed.

PENDING



Pump condition diagnosis

Requires Tier 2 data for a downhole card. Separately, the diagnostic classifier is calibrated on synthetic cards only and has not been scored against field cards with known outcomes — its labels are not yet reportable.

PENDING

6 Way forward

What each additional input unlocks, ordered by how much it changes the answer.

6.1 Stage A — card integrity

REQUIRED	
Rod tally or last workover ticket	outstanding – card implies 7/8 in
Load-cell make, model, calibration date	outstanding
Was the cell zeroed with rods hanging?	outstanding
Runtime, hours per day	outstanding

RETURNS	
Resolved load datum	absolute loads usable
Confirmed rod string	corrects every load-based result
Honest production check	efficiency with runtime
Effort	three questions

6.2 Stage B — downhole card and diagnosis

REQUIRED	
Tubing ID and anchor status	outstanding
Fluid level above pump	outstanding
Oil gravity and B_o	outstanding
Viscosity or bottomhole temperature	outstanding
RETURNS	
Calculated downhole pump card	the open question
Pump fillage	measured, not assumed
Condition separation	pump-off / gas / wear
Solver status	ready – not blocked on code

6.3 Stage C — mechanical loading

Requires the pumping unit's API designation and its API 11E geometry sheet, plus the motor nameplate. Returns gearbox torque against the unit's rating, counterbalance condition, rod loading and buckling, and power consumption.

Note the C-66 is an Arrow Engine natural-gas prime mover, not a gearbox rating, so no torque limit is available.

6.4 Better data beats more data

A raw controller export — .dyn or timestamped CSV — outranks several Stage B items. Timing read off a position-axis card carries ± 0.11 to ± 0.15 s at ± 1.5 in digitisation error, because rod velocity vanishes at both stroke ends. At that uncertainty the apparent peak spacings of 1.24 s and 0.95 s are not distinguishable from each other, nor from the predicted 1.03 s.

6.5 Diagnostic calibration — open initiative

The diagnostic layer requires field cards with known outcomes to become trustworthy. A card alone contributes little; the pulling or workover report is the informative half. Highest value: cards paired with what was found on the pull, before-and-after pairs on one well, and confirmed healthy cards — the last being scarcer than failures and the baseline everything else is measured against. Contributed data is analysed and returned at no charge, the work is open source with no licence dependency, and early contributors are supported for life.

7 References & provenance

Methods, data sources, traceability.

- R1** API RP 11L — Recommended Practice for Design Calculations for Sucker Rod Pumping Systems.
- R2** API 11E — Specification for Pumping Units.
- R3** Everitt, T.A. and Jennings, J.W., “An Improved Finite-Difference Calculation of Downhole Dynamometer Cards for Sucker-Rod Pumps”, SPE 18189, SPE Production Engineering.
- R4** Gibbs, S.G., “Predicting the Behavior of Sucker-Rod Pumping Systems”, JPT, 1963.
- R5** Rowlan, O.L. (Echometer), “Over Travel Occurs on Both the Upstroke and Down Stroke”, Sucker Rod Pumping Workshop / SWPSC.

- R6 Source data: operator-reported well parameters and surface dynamometer card, Collide “Dynamometer Discussions” thread, July 2026. Rod string confirmed by the operator 27 July 2026.
- R7 Implementation and validation data, public: github.com/vamseeachanta/digitalmodel — API RP 11L rod-pump module and SPE 18189 solver, with regression tests against five anonymised field wells.

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Prepared by AceEngineer Artificial Lift.

PROVENANCE

- Teal — validated
- Amber — analytical, assumptions carried
- Muted — pending data